

OVERTHEWIRE LEVEL-BY- LEVEL BREAKDOWN

INTRODUCTION

OverTheWire is a website that offers free online games to help people learn cybersecurity. These games are called "wargames" and they are designed to teach skills like Linux commands, networking, and basic hacking techniques. Each game has different levels, and players solve challenges to move from one level to the next. OverTheWire is useful for beginners who want hands-on practice and for students who want to improve their problem-solving skills in cybersecurity.

OBJECTIVES

- Teach cybersecurity basics through hands-on practice.
- Help learners understand Linux commands, file systems, and networking.
- Improve problem-solving skills by solving real-world style challenges.
- Prepare students for ethical hacking and penetration testing tasks.
- Build confidence in using command-line tools and thinking like a hacker.

FEATURES

- Terminal-based challenges that simulate real cybersecurity scenarios.
- Structured wargames with increasing difficulty levels.
- Covers key topics like file permissions, SSH, scripting, and privilege escalation.
- Multiple games available for different skill levels.
- No installation required—users connect via SSH to start solving.
- Promotes independent learning by encouraging research and exploration.

SOLVING

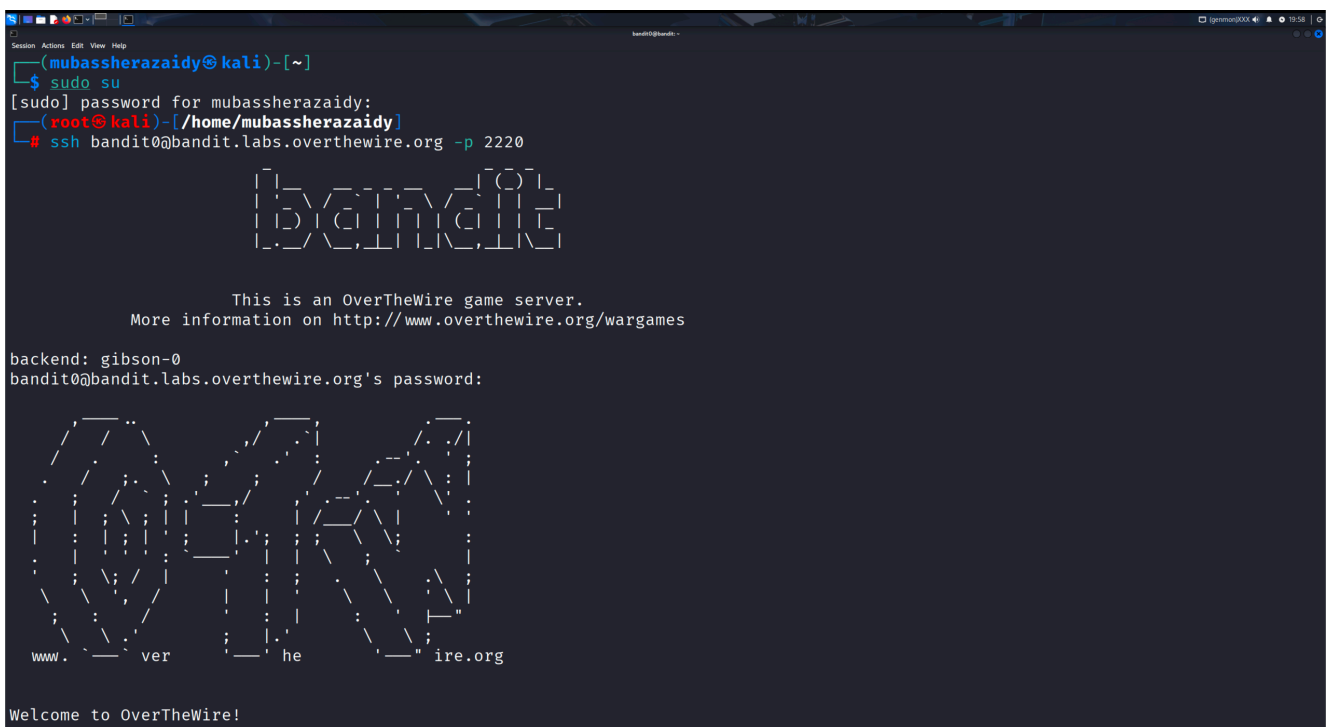
LEVEL 0 : The goal of this level is for you to log into the game using SSH. The host to which you need to connect is `bandit.labs.overthewire.org`, on port 2220. The username is `bandit0` and the password is `bandit0`.

Given: IP of host - `bandit.labs.overthewire.org`

Username - `bandit0`

Password - `bandit0`

Port - 2220



```
Session Actions Edit View Help
(mubassherazaidy@kali)-[~]
└─$ sudo su
[sudo] password for mubassherazaidy:
(mubassherazaidy@kali)~$ ssh bandit0@bandit.labs.overthewire.org -p 2220

      _ _ _ _ _
     / /   / /
    / /   / /
   / /   / /
  / /   / /
 / /   / /
/_/_/_/_/_

      This is an OverTheWire game server.
      More information on http://www.overthewire.org/wargames

backend: gibbon-0
bandit0@bandit.labs.overthewire.org's password:

  _ _ _ _ _
 / /   / /
/_/_/_/_/_

www.   ver   he   ire.org

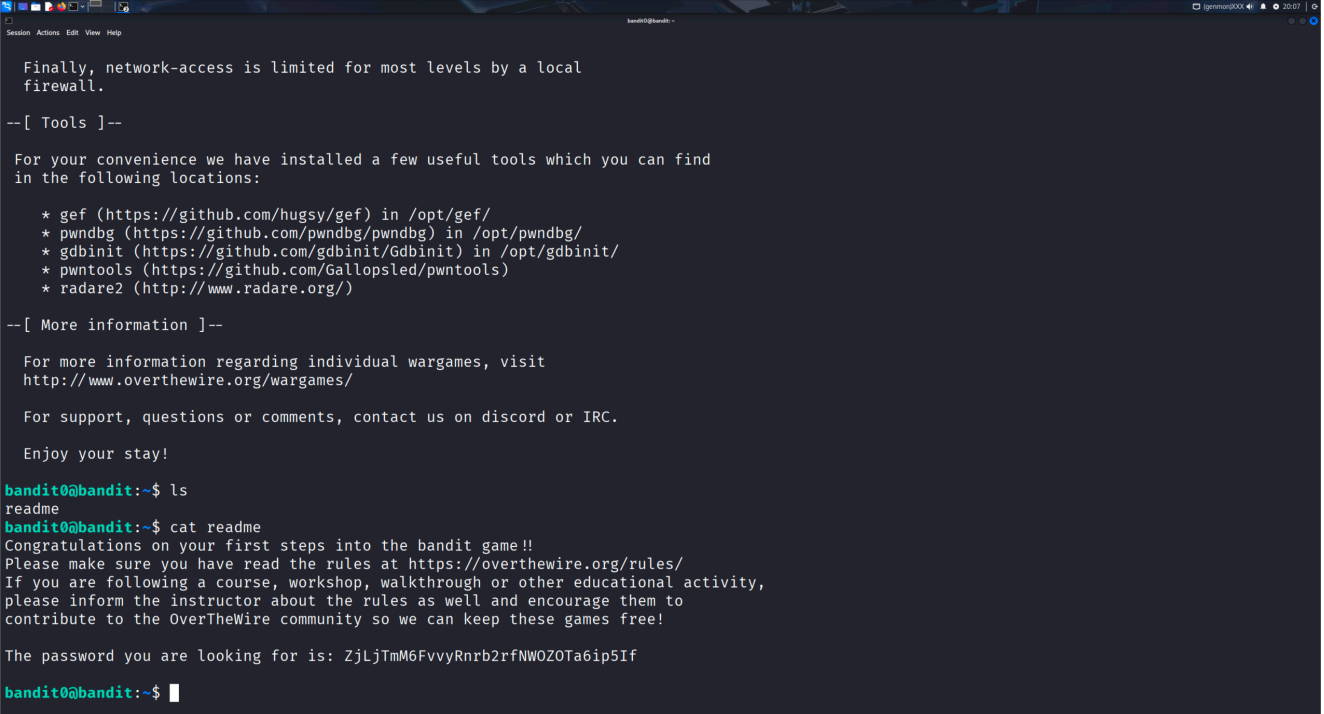
Welcome to OverTheWire!
```

LEVEL 1 : The password for the next level is stored in a file called readme located in the home directory. Use this password to log into bandit1 using SSH. Whenever you find a password for a level, use SSH (on port 2220) to log into that level and continue the game.

Given : file name - "readme" in home directory

Command to use: `ls` - Lists all files/directory in the current directory

`cat readme` - Displays the content of file name readme



```
Finally, network-access is limited for most levels by a local
firewall.

--[ Tools ]--

For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/Gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
* radare2 (http://www.radare.org/)

--[ More information ]--

For more information regarding individual wargames, visit
http://www.overthewire.org/wargames/

For support, questions or comments, contact us on discord or IRC.

Enjoy your stay!

bandit0@bandit:~$ ls
readme
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5If

bandit0@bandit:~$
```

Password found - ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5If

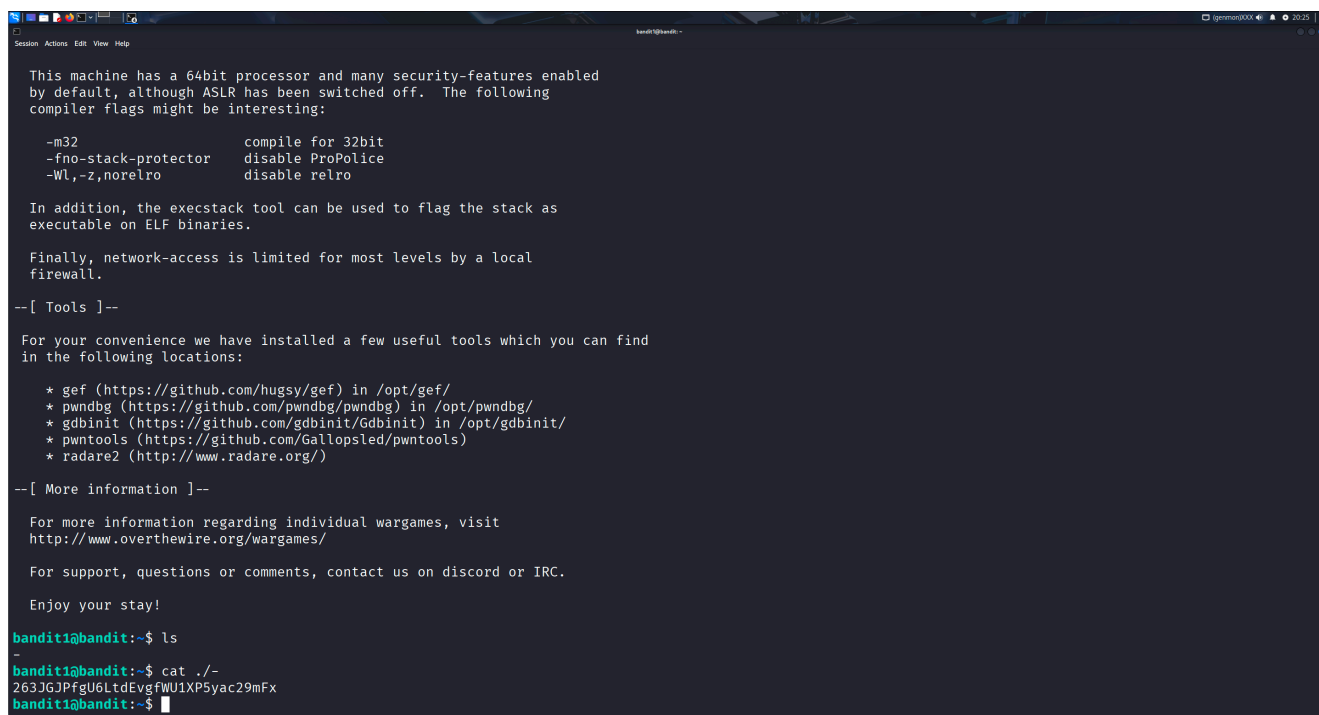
LEVEL 2 : The password for the next level is stored in a file called - located in the home directory

Given : Name of file : " - "

Command to use : `ls` : Lists all files in the current directory

`cat ./- :` to show content in file

- `./` means "current directory"
- `-` is the actual filename
- Together, `./-` tells `cat` to treat `-` as a literal file name, not a command option or input stream.



```
bandit@bandit:~$ cat ./-
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

-m32                compile for 32bit
-fno-stack-protector disable ProPolice
-Wl,-z,norelro       disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
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bandit@bandit:~$ ls
-
bandit@bandit:~$ cat ./-
263JGJPfgU6LtdEvvgfWU1XP5yac29mFx
bandit@bandit:~$
```

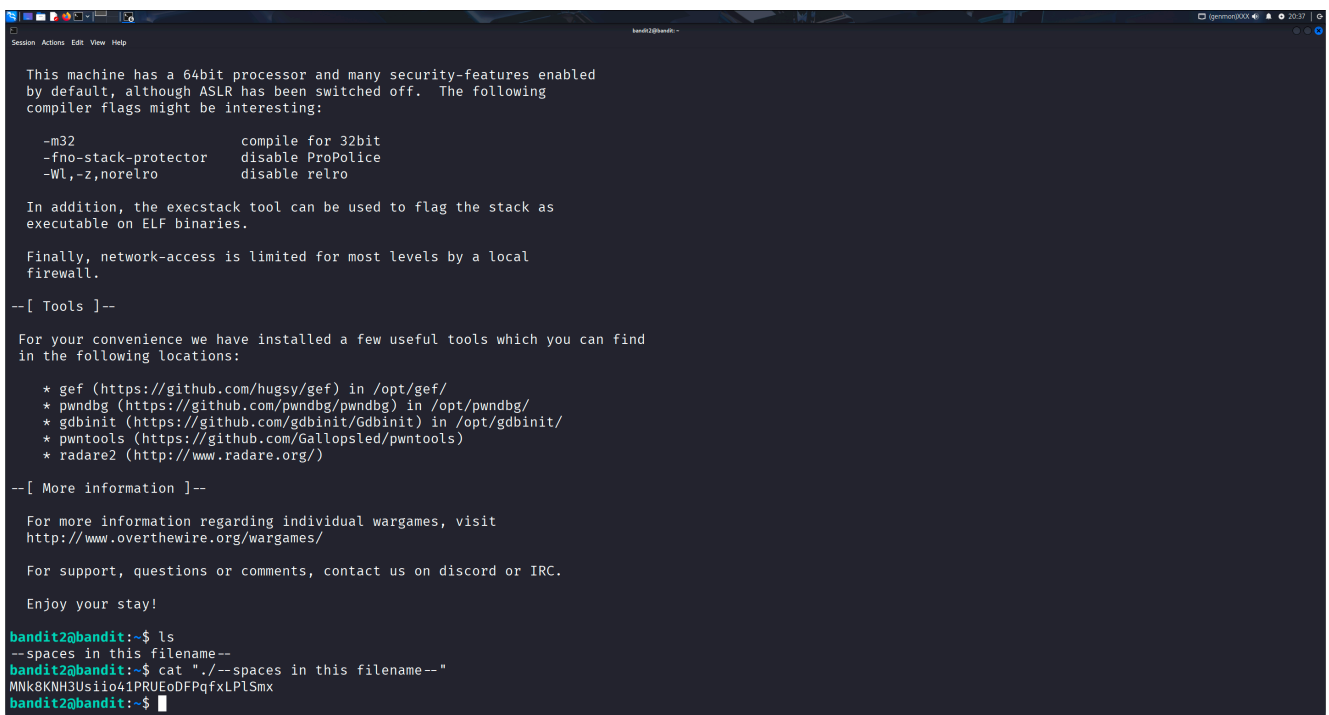
Password found - 263JGJPfgU6LtdEvvgfWU1XP5yac29mFx

LEVEL 3 : The password for the next level is stored in a file called `--spaces in this filename--` located in the home directory

Given : Name of file : `"--spaces in this filename--"` in home directory

Commands to use : `ls` : Lists all files in the current directory

`cat "--spaces in this filename--"` : to show content in file



```
bandit2@bandit:~$ cat /dev/null
This machine has a 64bit processor and many security-features enabled
by default, although ASLR has been switched off. The following
compiler flags might be interesting:

-m32             compile for 32bit
-fno-stack-protector  disable ProPolice
-Wl,-z,norelro    disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

Finally, network-access is limited for most levels by a local
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bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ cat "--spaces in this filename--"
MNk8KNH3Usiio41PRUEoDFPqfxLPISmx
bandit2@bandit:~$
```

Password found - MNk8KNH3Usiio41PRUEoDFPqfxLPISmx

LEVEL 4 : The password for the next level is stored in a hidden file in the inhere directory.

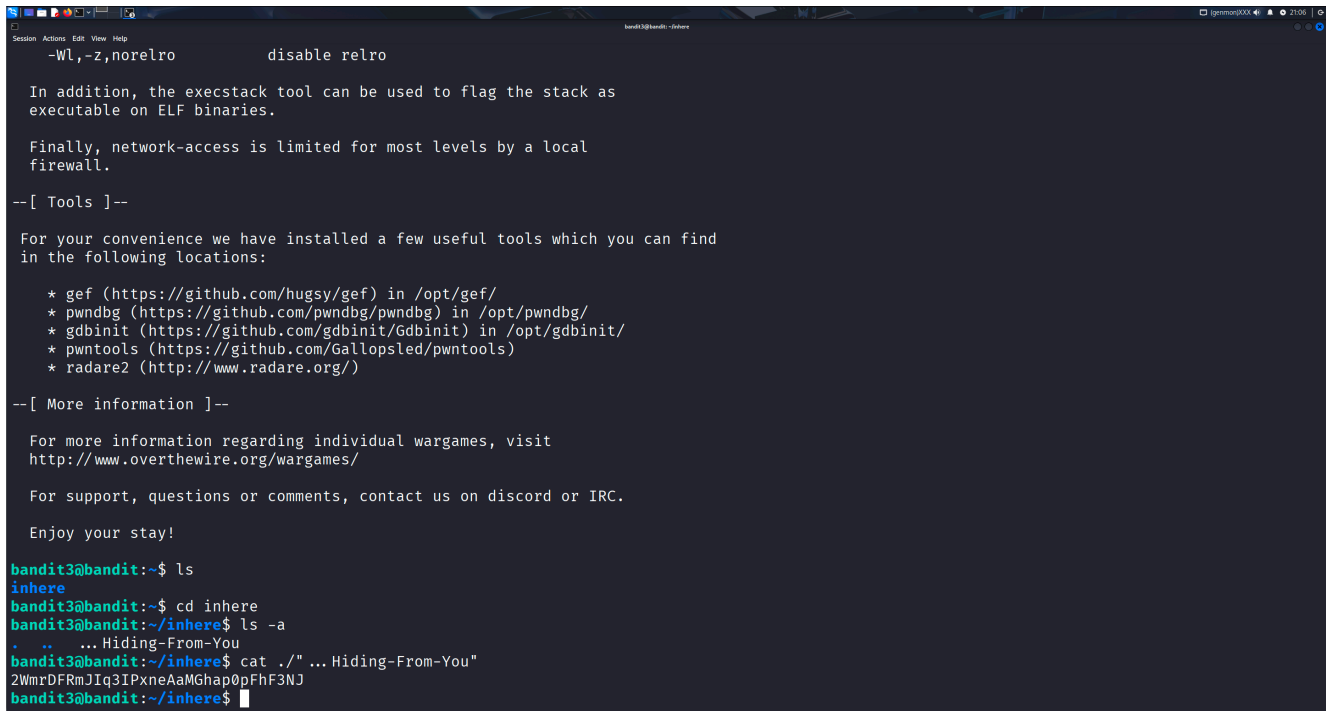
Given : Name of directory : inhere

Command to use : `ls` : Lists all files/directory in the current directory

`cd inhere` - to change the directory from the current to inhere.

`ls -a` : shows all files, including hidden ones

`cat "...Hiding-From-You"` : to show content in the file



```
bandit3@bandit:~$ ls
-Wl,-z,norelro      disable relro

In addition, the execstack tool can be used to flag the stack as
executable on ELF binaries.

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firewall.

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bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls -a
.  ..  ... Hiding-From-You
bandit3@bandit:~/inhere$ cat ./"... Hiding-From-You"
2WmrDFRmJIq3IPxneAaMGhapOpFhF3NJ
bandit3@bandit:~/inhere$
```

Password found - 2WmrDFRmJIq3IPxneAaMGhapOpFhF3NJ

LEVEL 5 : The password for the next level is stored in the only human-readable file in the inhere directory.

Given: Name of the directory : inhere

Commands to use : `ls` : Lists all files/directory in the current directory
`cd inhere` - to change the directory from the current to inhere.
`cat ./"-file07"` : to show content in the file

```
bandit4@bandit:~$ cat /dev/null
For your convenience we have installed a few useful tools which you can find
in the following locations:

* gef (https://github.com/hugsy/gef) in /opt/gef/
* pwndbg (https://github.com/pwndbg/pwndbg) in /opt/pwndbg/
* gdbinit (https://github.com/gdbinit/gdbinit) in /opt/gdbinit/
* pwntools (https://github.com/Gallopsled/pwntools)
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bandit4@bandit:~$
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls
-file00 -file01 -file02 -file03 -file04 -file05 -file06 -file07 -file08 -file09
bandit4@bandit:~/inhere$ cat ./"-file00"
***w***f***y***k***K***S***v***m***/*DGo**XI'bandit4@bandit:~/inhere$ cat ./"-file01"
*v***`Rg.*v0>0h      *]r*p*i*L_ *Lbandit4@bandit:~/inhere$ cat ./"-file02"
****J***x"*.x*****l***x***mbandit4@bandit:~/inhere$ cat ./"-file03"
|*g@*TU***uhjaWi***dS>*G***ycbandit4@bandit:~/inhere$ cat ./"-file04"
YQ*P***P***;***z*[*;***l***ë***bandit4@bandit:~/inhere$ cat ./"-file05"
*****xv***y*0lkG*N***J4"***RG*bandit4@bandit:~/inhere$ cat ./"-file06"
1D***d
    ***M*1***1***nq*)*йbandit4@bandit:~/inhere$ cat ./"-file07"
4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw
bandit4@bandit:~/inhere$
```

Password found - 4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw